

UDDHAV BHOPLE

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SUMMARY

AI engineer and B.Tech CSE student building LLM agents, RAG systems, and production-style back-end APIs with Python, FastAPI, PostgreSQL, Docker, and cloud (GCP, AWS, Azure), with a focus on guardrails, testing, and reliability.

TECHNICAL SKILLS

AI / LLM: LLM Agents, RAG, Prompt Engineering, LangChain, Vertex AI, Generative AI, NLP

Machine Learning: TensorFlow, scikit-learn, NumPy, Pandas

Back-end: Python, FastAPI, Flask, Node.js, Express, REST APIs, OpenAPI/Swagger, JWT, React, Next.js

Cloud / DevOps: GCP, AWS, Docker, Docker Compose, GitHub Actions, CI/CD, Linux, Git

Databases: PostgreSQL, MongoDB, MySQL, SQLite

Languages & Testing: Python, TypeScript, JavaScript, C, SQL, pytest

EXPERIENCE

FlyRank AI

Jul 2026 – Aug 2026

Backend AI Engineering Intern (Virtual Internship Program)

Remote

- Built a REST API in **Python/FastAPI** with full CRUD, input validation, correct HTTP status codes (200/201/204/400/404), and auto-generated **OpenAPI/Swagger** docs; published to GitHub with staged commits and a documented README.
- Training in the Backend AI Engineering track: REST design, databases, LLM integration, and **GCP** deployment; [add LangChain / Vertex AI work, or delete].

PrivGPT Studio

Jan 2026 – Apr 2026

Open-Source Contributor

- Contributed to a privacy-first AI workspace supporting local and cloud **LLMs** with offline fallback (TypeScript, Python, Next.js, Flask, MongoDB).
- Implemented multimodal cross-referencing across PDFs, video, images, and audio (with transcription) to improve contextual accuracy.

EduSkills

Apr 2025 – Jun 2025

AI-ML Virtual Intern

- Built a **machine learning** pipeline for treatment and diet recommendations using Python, TensorFlow, and Flask on a real-world dataset.
- Integrated Firebase authentication, Twilio, and Cloudinary with 98% service reliability.

PROJECTS

Recover.ai: Agentic Payment Recovery | Python, FastAPI, PostgreSQL, React, Docker, LLM, Razorpay API

julay 2026

- Built an AI agent that detects failed Razorpay subscription payments through signature-validated, idempotent webhooks and runs a **Diagnose** → **Decide** → **Guardrail** → **Act** pipeline with bounded recovery actions (retry or payment link).
- Added a deterministic guardrail so ambiguous LLM diagnoses cannot bypass policy; every action is written to a PostgreSQL audit trail before any external payment call.
- Shipped a React dashboard, delayed-retry worker, Alembic migrations, Docker Compose setup, and pytest/Ruff checks; a simulated 60-case batch recovered 14 cases (23.3%, Rs 10,886) and kept 46 exceptions for human review.

Autonomous Coding Agent | Python, Pydantic, pytest, Docker

2026

- Designing a research-driven AI software-engineering agent that works on existing repositories in a controlled workspace, planned across 15 phases covering context selection, tool permissions, verification, and evaluation.
- Implemented a secure workspace layer that rejects path traversal and symlink escapes and uses atomic writes (temp file + `os.replace`); covered by unit and integration tests and a Dockerized test run.

Bharat-Setu | Python, FastAPI, React, SQLAlchemy, JWT

Jan 2026 – Feb 2026

- Architected a zero-cost, pilot-ready platform with 3 modules (satellite-verified carbon credits, village digital twin, crisis detection), replacing 8 AWS services with open-source equivalents.
- Designed 6 SQLAlchemy models, 10 services, and 8 REST route groups; a fraud protocol auto-approves at <10% variance and freezes accounts at >30%. Shipped 76 unit tests across 4 suites, a React admin dashboard, and a mobile-first PWA with voice interface.

EDUCATION

D.Y. Patil University (DYPU)

B.Tech in Computer Engineering, Last Semester SGPA: 9.0

Pune, India

2024 – 2028

CERTIFICATIONS & LEADERSHIP

Certifications: Oracle Cloud AI Foundations, IBM Data Science, Saylor Academy AI, Microsoft Learn badges

Microsoft Learn Student Ambassador (Associate), Jan 2026 – Present: delivered technical sessions on Python, AI, and developer tools

Student Ambassador, Google (2025 – 2026) and Naukri